

Intrinsic p-type W-based transition metal dichalcogenide by substitutional Ta-doping

Cite as: Appl. Phys. Lett. **111**, 043502 (2017); <https://doi.org/10.1063/1.4995400>

Submitted: 01 May 2017 • Accepted: 09 July 2017 • Published Online: 24 July 2017

Yajun Fu, Mingsheng Long,  Anyuan Gao, et al.



View Online



Export Citation



CrossMark

ARTICLES YOU MAY BE INTERESTED IN

[p-type doping of MoS₂ thin films using Nb](#)

Applied Physics Letters **104**, 092104 (2014); <https://doi.org/10.1063/1.4867197>

[Band alignment of two-dimensional transition metal dichalcogenides: Application in tunnel field effect transistors](#)

Applied Physics Letters **103**, 053513 (2013); <https://doi.org/10.1063/1.4817409>

[Gate-tunable rectification inversion and photovoltaic detection in graphene/WSe₂ heterostructures](#)

Applied Physics Letters **108**, 223501 (2016); <https://doi.org/10.1063/1.4953152>

 QBLOX



1 qubit

Shorten Setup Time

Auto-Calibration

More Qubits

Fully-integrated

Quantum Control Stacks

Ultrastable DC to 18.5 GHz

Synchronized <<1 ns

Ultralow noise



100s qubits

visit our website >

Intrinsic p-type W-based transition metal dichalcogenide by substitutional Ta-doping

Yajun Fu,^{1,a)} Mingsheng Long,^{1,a)} Anyuan Gao,¹ Yu Wang,¹ Chen Pan,¹ Xiaowei Liu,¹ Junwen Zeng,¹ Kang Xu,¹ Lili Zhang,¹ Erfu Liu,¹ Weida Hu,² Xiaomu Wang,³ and Feng Miao^{1,b)}

¹National Laboratory of Solid State Microstructures, School of Physics, Collaborative Innovation Center of Advanced Microstructures, Nanjing University, Nanjing 210093, China

²National Laboratory for Infrared Physics, Shanghai Institute of Technical Physics, Chinese Academy of Sciences, Shanghai 200083, China

³School of Electronic Science and Technology, Nanjing University, Nanjing 210093, China

(Received 1 May 2017; accepted 9 July 2017; published online 24 July 2017)

Two-dimensional (2D) transition metal dichalcogenides (TMDs) have recently emerged as promising candidates for future electronics and optoelectronics. While most of TMDs are intrinsic n-type semiconductors due to electron donating which originates from chalcogen vacancies, obtaining intrinsic high-quality p-type semiconducting TMDs has been challenging. Here, we report an experimental approach to obtain intrinsic p-type Tungsten (W)-based TMDs by substitutional Ta-doping. The obtained few-layer Ta-doped WSe₂ (Ta_{0.01}W_{0.99}Se₂) field-effect transistor devices exhibit competitive p-type performances, including $\sim 10^6$ current on/off at room temperature. We also demonstrate high quality van der Waals (vdW) p-n heterojunctions based on Ta_{0.01}W_{0.99}Se₂/MoS₂ structure, which exhibit nearly ideal diode characteristics (with an ideality factor approaching 1 and a rectification ratio up to 1×10^5) and excellent photodetecting performance. Our study suggests that substitutional Ta-doping holds great promise to realize intrinsic p-type W-based TMDs for future electronic and photonic applications. *Published by AIP Publishing.*

[<http://dx.doi.org/10.1063/1.4995400>]

Since the recent emergence of transition metal dichalcogenides (TMDs), tremendous efforts have been involved to explore their interesting physical phenomena and potential applications.^{1–6} Due to their moderate band gap and strong light-matter coupling,^{7,8} two-dimensional (2D) TMDs are top material candidates for the next generation electronic and photonic devices.^{9–11} Like traditional complementary metal-oxide-semiconductor (CMOS) electronics, both p- and n-type TMDs are indispensable for realizing complementary device applications. However, most of semiconducting TMDs are intrinsic n-type originating from chalcogen vacancies. Although engineering extrinsic factors could help achieving p-type behaviors, such as metal work-function engineering,¹² ionic liquid gating,^{13,14} and chemical doping,¹⁵ obtaining intrinsic high quality p-type semiconducting TMDs has been challenging.

On the other hand, as a representative TMD material, tungsten diselenide (WSe₂) could be a good candidate to realize an intrinsic p-type semiconductor due to the observed ambipolar behavior^{16,17} and its relatively high hole mobility which benefits the device performances.^{12,18,19} Moreover, due to its unique electronic band structure and strong spin-orbit interaction (originating from W as a heavy 5d element), the spin degeneracy near the valence band edge of WSe₂ is split by ~ 400 meV, much stronger than that of MoS₂ (~ 160 meV).^{20,21} This results in many intriguing physical phenomena, such as valley pseudospins,²² electric-field-driven Zeeman-type spin splitting,²³ and negative electronic

compressibility.^{24,25} Therefore, considering the small effective hole mass ($0.45 m_0$),²⁴ high hole mobility, as well as the unique band structure, an intrinsic high-quality p-type WSe₂ could provide an excellent platform for exploring interesting physics and device applications.

In this work, we report the realization of p-type WSe₂ (Ta_{0.01}W_{0.99}Se₂) by substitutional Ta-doping. We perform various material characterizations with results indicating doping of Ta element. The few-layer Ta_{0.01}W_{0.99}Se₂ field-effect transistors (FETs) show competitive p-type performances including high current on/off ratio up to 10^6 . We further fabricate vdW p-n heterojunctions by vertically stacking Ta_{0.01}W_{0.99}Se₂ and few-layer MoS₂. Nearly ideal diode behaviors (with an ideality factor approaching 1 and a rectification ratio of 1×10^5) and high efficiency photoresponse were observed, suggesting high quality of our intrinsic p-type WSe₂ samples. Our results pave the way for future electronic and optoelectronic applications based on complementary TMD materials.

Elemental doping is an efficient and widely adopted method to adjust both carrier type and carrier density in traditional semiconductors. Recently, elemental doping method has been utilized to modulate the carrier type of Molybdenum (Mo)-based TMDs, resulting in intrinsic p-type characteristics.^{26,27} A feasible scenario to achieve p-type WSe₂ by element doping is to replace certain W atoms (six valence electrons) by Tantalum (Ta) atoms with five valence electrons and matched atomic radius.

Figure 1(a) schematically shows the crystalline structure of Ta-doped WSe₂, where Ta atoms replace W atoms randomly. The Ta-doped WSe₂ (nominal value of Ta in atomic

^{a)}Y. Fu and M. Long contributed equally to this work.

^{b)}Author to whom correspondence should be addressed: miao@nju.edu.cn.

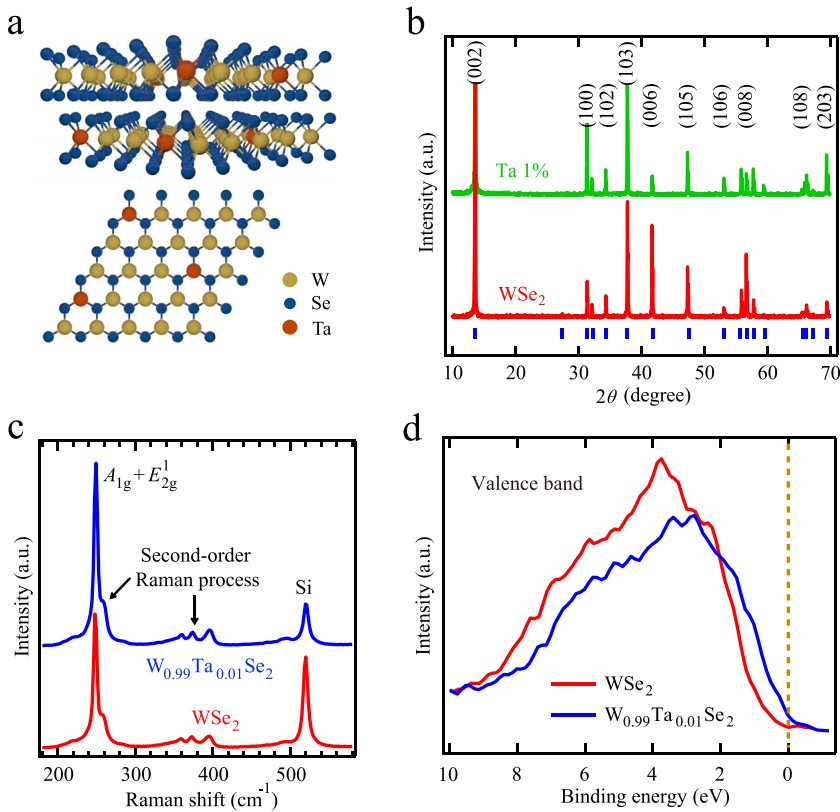


FIG. 1. Characterization of Ta-doped WSe₂. (a) Schematic diagrams of the Ta-doped WSe₂. The dopant Ta atoms randomly replace W atoms results in a hole doped WSe₂. (b) XRD patterns for Ta-doped and pristine WSe₂. The little blue sticks at the bottom present the peak position of WSe₂ crystal from the standard PDF card #38-1388. (c) Raman spectra of exfoliated monolayer Ta-doped and pristine WSe₂ flakes. (d) Valence band spectra of Ta-doped and pristine WSe₂ measured by XPS. The dashed line marks the location of Fermi level.

ratio is $\sim 1\%$) bulk single crystals were grown by chemical vapor transport (CVT) method²⁸⁻³⁰ (see details in experimental section). We identified the crystal phase by utilizing X-ray diffraction (XRD) in a powder mode [Fig. 1(b)]. The peak positions for Ta-doped and pristine WSe₂ are almost identical and consistent with the standard PDF card #38-1388,

indicating that good crystallinity (2H phase) was maintained in the Ta-doped WSe₂. According to the XRD diffraction angle data, the lattice constant along c axis was calculated to be 12.977 Å and 12.960 Å for pristine and Ta-doped WSe₂, respectively. The in-plane constant a maintains almost the same with results calculated to be 3.289 Å and 3.291 Å for

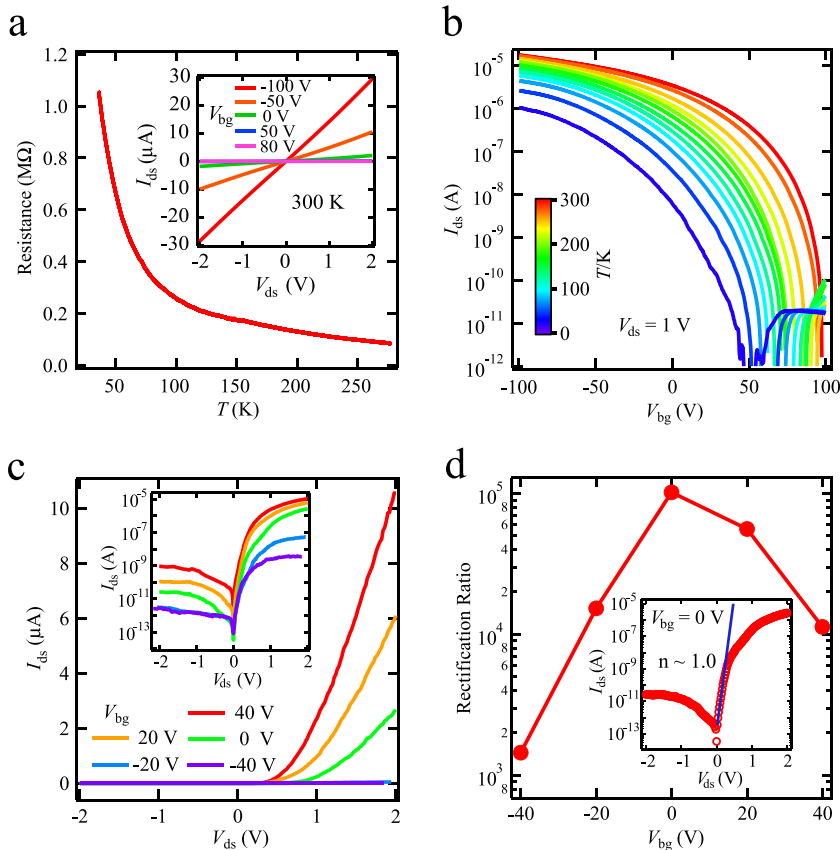


FIG. 2. FETs and p-n heterojunctions based on Ta_{0.01}W_{0.99}Se₂. (a) Temperature dependent resistance of Ta_{0.01}W_{0.99}Se₂ measured by four terminal geometry. Inset shows the I_{ds} - V_{ds} curves for different back gate biases at 300 K, the linear character suggests good contact. (b) The temperature dependent Logarithmic-scale transfer curves of a Ta_{0.01}W_{0.99}Se₂ FET. The temperature changed from 25 K to 300 K with a step of 25 K. (c) Gate tunable rectifying behavior of a typical p-n heterojunction build by vertical stacking few-layer Ta_{0.01}W_{0.99}Se₂ and MoS₂. Inset: Semilogarithmic plot of the I_{ds} - V_{ds} curves. (d) The rectification ratio [extracted from the inset of (c)] as a function of gate bias. Inset: Semilogarithmic plot of the I - V curve for $V_{bg} = 0$ V. The blue line shows linear fit to the diode equation and obtained an ideality factor of $n \sim 1$.

pristine and Ta-doped WSe₂, respectively. Such results suggest that certain W atoms are replaced by Ta atoms, in good consistency with prior work for Nb-doped MoSe₂.²⁷ The Raman spectra of two different types of WSe₂ (monolayer flakes) are shown in Fig. 1(c), exhibiting similar characteristics. The main peak around 250 cm⁻¹ is associated with both A_{1g} and E_{2g}¹ modes (with small frequency difference). The small features located around 260, 360, 373, and 395 cm⁻¹ correspond to the second-order Raman processes.^{31,32} We noticed that the main peak (A_{1g} and E_{2g}¹ modes) of Ta-doped WSe₂ is slightly blueshifted with ~ 1.4 cm⁻¹, which could be induced by hole doping induced phonon hardening.³³

We further characterized the Ta element doping by utilizing X-ray photoelectron spectroscopy (XPS) and Energy Dispersive X-ray spectroscopy (EDX). XPS spectra clearly show the existence of Ta element in the doped WSe₂ in spite of weak feature of the Ta 4f peak (see Fig. S1 in the supplementary material). According to the XPS data shown in Fig. 1(d), the Fermi level of Ta-doped WSe₂ is downshifted by ~ 0.4 eV while the shape feature of the valence band keeps consistent, indicating that the Fermi level of the doped samples is much closer to the valence band maximum. At the same time, W 4f and Se 3d peaks in XPS spectra of Ta-doped WSe₂ are upshifted by ~ 0.4 eV as well (see Fig. S1 in the supplementary material). Although the peak feature of Ta 4f in XPS data is too weak for the calculation of the atomic ratio for each element, a value of ~ 0.7 at. % Ta was obtained for Ta_{0.01}W_{0.99}Se₂ samples according to the EDX data (see Fig. S2 in the supplementary material), close to the nominal elemental composition. Overall, these results clearly demonstrate that certain W atoms in WSe₂ single crystals are substituted by Ta atoms, resulting in intrinsic hole doping.

We now focus on the electronic properties of the few-layer Ta_{0.01}W_{0.99}Se₂. Thin flakes (thickness below 5 nm) were exfoliated onto degenerately doped Si wafers covered with 300 nm SiO₂ for device fabrications. For a typical few-layer Ta_{0.01}W_{0.99}Se₂ device (3.4 nm), the source-drain current I_{ds} exhibits good linear dependence on the bias V_{ds} (within the range of ± 2 V) under various gate voltages V_{bg} [inset of Fig. 2(a)], which implies the ohmic contact behavior for the hole transport at room temperature (300 K). The temperature dependent four-terminal resistance measurements were carried out, where the resistance increases monotonically with decreasing temperature, indicating a semiconducting transport behavior [as shown in Fig. 2(a)]. Additionally, the FET transfer curves measured from 25 to 300 K show excellent p-type FET characteristics when the back gate swept between -100 and $+100$ V [as shown in Fig. 2(b)]. The current on/off ratio exceeded 10^6 at room temperature, which fulfills the requirement of logic applications. We noticed that a weak n-type current for high positive gate biases appeared at lower temperatures. At room temperature, the subthreshold swing is about 1 V dec⁻¹ for 300 nm back gate dielectric (see Fig. S3 in the supplementary material), which could be further improved by utilizing thinner high- κ dielectric materials. The field-effect mobility was extracted from the linear regime of the transfer curves (see Fig. S3 in the supplementary material), which reached ~ 16.5 cm² V⁻¹ s⁻¹ at room temperature. According to the temperature dependent transfer characteristics [Fig. 2(b)], we found the data fit well to

a hopping transport model (see Fig. S4 in the supplementary material).³⁴ Similar to previous studies in MoS₂, the charge transport in Ta_{0.01}W_{0.99}Se₂ is thermally activated and switched from the nearest hopping to the variable range hopping at lower temperature regime.^{34,35} We also performed control experiments to study the FET properties of pristine WSe₂ (without Ta dopants) few layer flakes, Ta_{0.005}W_{0.995}Se₂ thin flakes with different thickness and Ta_{0.005}W_{0.995}Se₂ on h-BN substrate (see Figs. S5–S7 in the supplementary material), where the pristine WSe₂ shows ambipolar behavior, and Ta_{0.005}W_{0.995}Se₂ samples exhibit stable p-type conductance.

The demonstration of intrinsic p-type WSe₂ makes it feasible to realize high-quality p-n heterojunctions. Figure 2(c) shows the typical gate tunable rectifying behavior at room

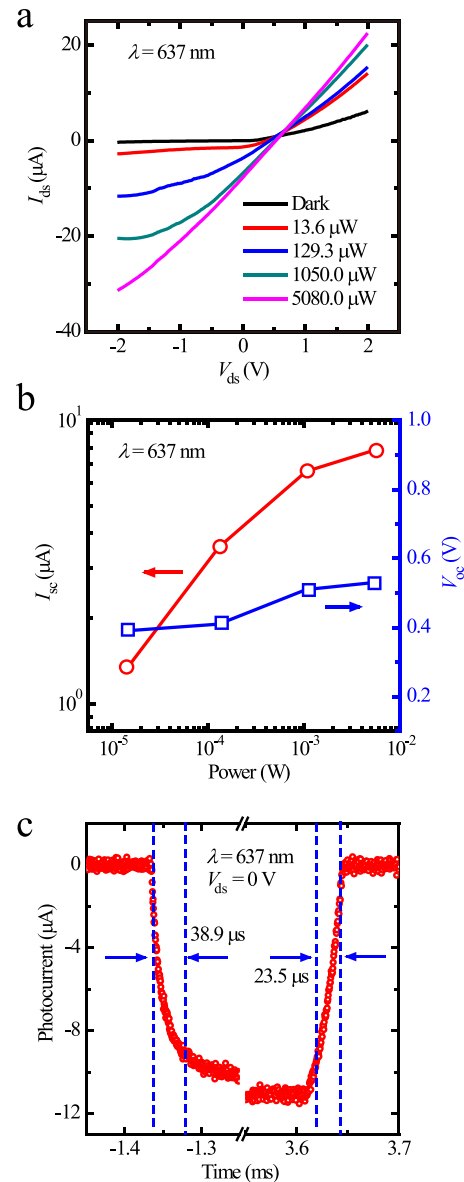


FIG. 3. Photoresponse of a typical Ta_{0.01}W_{0.99}Se₂/MoS₂ p-n heterojunction. (a) I_{ds} - V_{ds} curves measured by exposing the p-n diode to a focused laser beam (637 nm) with different power intensity. (b) Short-circuit current (I_{sc}) and open-circuit voltage (V_{oc}) extracted from (a) as a function of various incident power intensity. (c) Fast photoresponse of the p-n junction measured by using a laser (637 nm) of 1000 Hz frequency. Here the rise/fall time was defined as the photocurrent increased/decreased from 10/90% to 90/10% of the stable photocurrent.

temperature for a vertical stacked $\text{Ta}_{0.01}\text{W}_{0.99}\text{Se}_2$ (2.3 nm) and n-MoS_2 (2.7 nm) vdW heterojunction (with Ti/Au electrodes), in which the forward current at positive source-drain voltage increases with increasing V_{bg} . The rectification ratio, which is defined as the forward/reverse current ratio at the same source-drain bias, exceeded 10^3 for all measured gate voltages [$V_{\text{ds}} = \pm 2$ V, as shown in the inset of Fig. 2(c)]. The extracted rectification ratios ($V_{\text{ds}} = \pm 2$ V) at different gate biases are shown in Fig. 2(d), where the highest rectification ratio achieved is up to 1×10^5 at $V_{\text{bg}} = 0$ V. In addition, we fitted the rectification curves of the heterojunction device to the Shockley diode equation³⁶ and obtained an ideality factor (n) of ~ 1 [inset of Fig. 2(d)], which suggests an ideal diode behavior.

Due to the excellent diode behavior, the $\text{Ta}_{0.01}\text{W}_{0.99}\text{Se}_2/\text{MoS}_2$ p-n heterojunction could result in a pronounced photovoltaic effect. The photoresponse data of our p-n junction exposed to a focused laser beam with various excitation powers (at a fixed laser wavelength $\lambda = 637$ nm) are shown in Fig. 3(a). The extracted short circuit current (I_{sc}) and open circuit voltage (V_{oc}) are shown in Fig. 3(b). The highest V_{oc} obtained here is ~ 0.5 V. The extracted photoresponsivity (R) and external quantum efficiency (EQE) increase while decreasing the laser power and reach values of up to ~ 95 mA W^{-1} and $\sim 19\%$ ($P_{\text{Laser}} = 13.6$ μW), respectively (see Fig. S8 in the supplementary material). Moreover, similar photovoltaic behavior was also observed in vdW homojunction built by vertical stacking of Ta-doped WSe_2 and pristine WSe_2 (see Fig. S9 in the supplementary material).

The speed of photoresponse for photodetection devices is an important parameter to assess the device performance. A rise time of 38.9 μs and a fall time of 23.5 μs were observed [as displayed in Fig. 3(c)], which are almost three orders of magnitude faster than the double gated WSe_2 lateral p-n junction (~ 10 ms),³⁷ and comparable to the p-g-n heterostructure.³⁸ For TMDs van der Waals heterostructures, the ultrafast charge transfer and interlayer inelastic tunneling process promote the separation of electron-hole pairs at the interface.^{4,39} Therefore, taking into account the strong built-in electric field in vertical vdW heterojunctions, the high speed of photoresponse is attribute to the efficient charge separation at the vertical heterostructure interface.

We further carried out the photocurrent mapping measurements on the devices. Figure 4(a) shows the measured heterojunction, where the TMDs thin flakes are highlighted by dotted line with different colors, i.e., white and red colors represent $\text{Ta}_{0.01}\text{W}_{0.99}\text{Se}_2$ (2.9 nm) and MoS_2 (2.7 nm), respectively (see Fig. S10 in the supplementary material). The photocurrent mapping measurements were performed under ambient conditions, by using a 637 nm laser with power of ~ 269 μW . Figure 4(b) shows the photocurrent mapping at $V_{\text{ds}} = 0$ V and $V_{\text{bg}} = 0$ V, where the most pronounced photoresponse appears within the overlapped region of $\text{Ta}_{0.01}\text{W}_{0.99}\text{Se}_2$ and MoS_2 , originating from the formed atomically thin p-n junctions.

In summary, we have demonstrated the realization of intrinsic p-type W-based TMD by substitutional Ta-doping. The fabricated $\text{Ta}_{0.01}\text{W}_{0.99}\text{Se}_2$ FET devices exhibit competitive p-type performance, with current on/off ratio reached 10^6 . By vertical stacking $\text{Ta}_{0.01}\text{W}_{0.99}\text{Se}_2$ and few-layer MoS_2 , the formed vdW p-n heterojunctions show excellent diode and photoresponse characteristics. Our results enrich the understanding of substitutional doped p-type TMDs, and also provide a platform to explore interesting physics and device applications.

Ta-doped and pristine WSe_2 bulk single crystals were synthesized by using the chemical vapor transport (CVT) method with iodine as a transporting agent. Ta, W, and Se powders were stoichiometrically mixed with an atomic ratio of $\text{Ta}:\text{W}:\text{Se} = x:1-x:2$ ($x = 0, 0.005, 0.01, 0.02$). The powder mixtures were pressed in tablets and sealed in quartz tubes under vacuum (10^{-5} Torr), followed by two weeks heat treatment in a two zone furnace where the source and growth zones of the quartz tube were kept at 980 $^\circ\text{C}$ and 850 $^\circ\text{C}$, respectively. Then, the furnace was cooled down to room temperature within 5 h. The yielding single-crystals could be found in cold end of the quartz tube.

The crystal structure of Ta-doped WSe_2 flakes were determined by X-ray diffraction (X' TRA) with Cu $K\alpha$ radiation ($\lambda = 1.54059$ $\text{\AA}$). X-ray photoelectron spectroscopy (XPS, PHI 5000 VersaProbe) measurement was performed in ultra-high vacuum chamber ($\leq 6.7 \times 10^{-10}$ mbar), with an Al $K\alpha$ monochromatic photon source. The Energy Dispersive X-ray spectroscopy (EDX) measurement was carried out in a scanning electron microscope (SEM, S-3400N II) equipped with

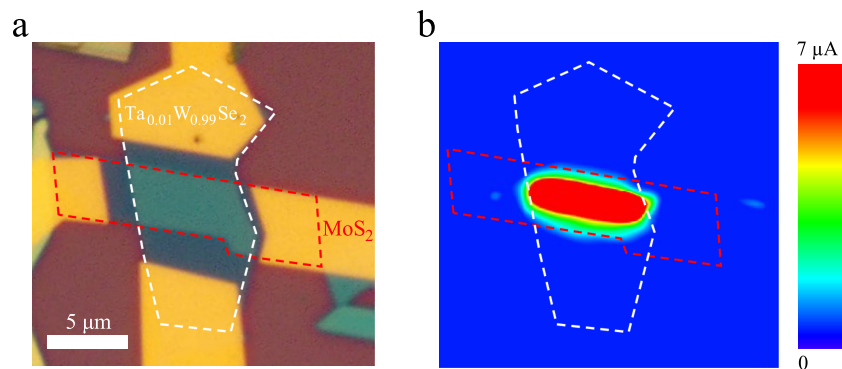


FIG. 4. Photocurrent mapping of a typical $\text{Ta}_{0.01}\text{W}_{0.99}\text{Se}_2/\text{MoS}_2$ p-n heterojunction. (a) Optical micrograph of the measured device, the scale bar is 5 μm . MoS_2 and Ta-doped WSe_2 are highlighted by red and white dashed lines, respectively. (b) Photocurrent mapping of the device shown in (a) at $V_{\text{ds}} = 0$ V and $V_{\text{bg}} = 0$ V. The measurements were performed under ambient conditions with a laser (637 nm) power intensity of 269 μW . The same dashed line design as in (a) are guides to the eye. The photocurrent mapping unambiguously shows the photoresponse originates from the overlapped region (junction region) of MoS_2 and $\text{Ta}_{0.01}\text{W}_{0.99}\text{Se}_2$.

an X-ray Energy Dispersive Detector (EX-250). The thicknesses of WSe₂ and MoS₂ thin flakes were determined by using a Bruker Multimode 8 atomic force microscope (AFM). Raman spectra measurements were carried out by utilizing the Raman spectrometer with a 514.5 nm wavelength laser (under ambient conditions).

The WSe₂ and MoS₂ thin flakes on degenerately doped silicon wafers (covered by 300 nm SiO₂) were fabricated by standard mechanical exfoliation method. The vertical heterojunctions (and homojunctions) were fabricated by using the method developed by Dean *et al.*⁴⁰ To fabricate electrical devices, electron-beam lithography process (FEI F50 with an NPGS pattern generation system) was used to design the electrodes patterns, and electron-beam evaporation was utilized to evaporate electrodes metals (typically 5 nm Ti/40 nm Au). The variable-temperature measurements were carried out in an Oxford Instruments TeslatronTM^{CF} cryostat. All the current signals were detected by Keithley 2636A dual channel digital source meter.

See [supplementary material](#) for additional experimental data of Ta-doped WSe₂.

M. L., Y. F., and F. M. conceived the project and designed the experiments. Y. F., M. L., Y. W., A. G., C. P., and P. W. performed device fabrication and characterization. M. L., Y. F., X. W., and F. M. performed data analysis and interpretation. Y. F., M. L., F. M., and X. W. co-wrote the paper, and all authors contributed to the discussion and preparation of the manuscript. This work was supported in part by the National Key Basic Research Program of China (2015CB921600, 2013CBA01603, 2013CB632700), the National Natural Science Foundation of China (11374142, 11322441, 61574076), the Natural Science Foundation of Jiangsu Province (BK20140017, BK20150055), the Fund of the Shanghai Science and Technology Foundation (14JC1406400), the Specialized Research Fund for the Doctoral Program of Higher Education (20130091120040), and Fundamental Research Funds for the Central Universities and the Collaborative Innovation Center of Advanced Microstructures.

The authors declare that they have no competing financial interests.

¹B. Radisavljevic, A. Radenovic, J. Brivio, I. V. Giacometti, and A. Kis, *Nat. Nanotechnol.* **6**, 147 (2011).

²Q. H. Wang, K. Kalantar-Zadeh, A. Kis, J. N. Coleman, and M. S. Strano, *Nat. Nanotechnol.* **7**, 699 (2012).

³K. F. Mak, K. L. McGill, J. Park, and P. L. McEuen, *Science* **344**, 1489 (2014).

⁴X. P. Hong, J. Kim, S. Shi, Y. Zhang, C. H. Jin, Y. H. Sun, S. Tongay, J. Q. Wu, Y. F. Zhang, and F. Wang, *Nat. Nanotechnol.* **9**, 682 (2014).

⁵D. B. Sarkar, X. J. Xie, W. Liu, W. Cao, J. H. Kang, Y. J. Gong, S. Kraemer, P. M. Ajayan, and K. Banerjee, *Nature* **526**, 91 (2015).

⁶E. F. Liu, Y. J. Fu, Y. J. Wang, Y. Q. Feng, H. M. Liu, X. A. Wan, W. Zhou, B. G. Wang, L. B. Shao, C. H. Ho, Y. S. Huang, Z. Y. Cao, L. G. Wang, A. D. Li, J. W. Zeng, F. Q. Song, X. R. Wang, H. T. Yuan, H. Y. Hwang, Y. Cui, F. Miao, and D. Y. Xing, *Nat. Commun.* **6**, 6991 (2015).

⁷L. Britnell, R. M. Ribeiro, A. Eckmann, R. Jalil, B. D. Belle, A. Mishchenko, Y. Kim, R. V. Gorbachev, T. Georgiou, S. V. Morozov, A. N. Grigorenko, A. K. Geim, C. Casiraghi, A. H. Castro Neto, and K. S. Novoselov, *Science* **340**, 1311 (2013).

⁸X. Z. Liu, T. Galfsky, Z. Sun, F. N. Xia, E. Lin, Y. Lee, S. Kéna-Cohen, and V. M. Menon, *Nat. Photonics* **9**, 30 (2015).

⁹A. K. Geim and I. V. Grigorieva, *Nature* **499**, 419 (2013).

¹⁰S. B. Desai, S. R. Madhupathy, A. B. Sachid, J. P. Llinas, Q. X. Wang, G. H. Ahn, G. Pitner, M. J. Kim, J. Bokor, C. M. Hu, H. S. P. Wong, and A. Javey, *Science* **354**, 99 (2016).

¹¹K. S. Novoselov, A. Mishchenko, A. Carvalho, and A. C. Neto, *Science* **353**, aac9439 (2016).

¹²H. Fang, S. Chuang, T. C. Chang, K. Takei, T. Takahashi, and A. Javey, *Nano Lett.* **12**, 3788 (2012).

¹³H. Chuang, X. B. Tan, N. J. Ghimire, M. M. Perera, B. Chamlagain, M. M. Cheng, J. Q. Yan, D. Mandrus, D. Tomañek, and Z. X. Zhou, *Nano Lett.* **14**, 3594 (2014).

¹⁴Y. J. Zhang, J. T. Ye, Y. Yomogida, T. Takenobu, and Y. Iwasa, *Nano Lett.* **13**, 3023 (2013).

¹⁵X. Liu, D. Qu, J. Ryu, F. Ahmed, Z. Yang, D. Lee, and W. J. Yoo, *Adv. Mater.* **28**, 2345 (2016).

¹⁶S. Das and J. Appenzeller, *Appl. Phys. Lett.* **103**, 103501 (2013).

¹⁷A. Allain and A. Kis, *ACS Nano* **8**, 7180 (2014).

¹⁸H. L. Zhou, C. Wang, J. C. Shaw, R. Cheng, Y. Chen, X. Q. Huang, Y. Liu, N. O. Weiss, Z. Y. Lin, Y. Huang, and X. F. Duan, *Nano Lett.* **15**, 709 (2015).

¹⁹V. Podzorov, M. E. Gershenson, C. Kloc, R. Zeis, and E. Bucher, *Appl. Phys. Lett.* **84**, 3301 (2004).

²⁰K. F. Mak, K. L. He, J. Shan, and T. F. Heinz, *Nat. Nanotechnol.* **7**, 494 (2012).

²¹X. D. Xu, W. Yao, D. Xiao, and T. F. Heinz, *Nat. Phys.* **10**, 343 (2014).

²²G. Aivazian, Z. R. Gong, A. M. Jones, R. Chu, J. Q. Yan, D. G. Mandrus, C. W. Zhang, D. Cobden, W. Yao, and X. D. Xu, *Nat. Phys.* **11**, 148 (2015).

²³H. T. Yuan, M. S. Bahramy, K. Morimoto, S. F. Wu, K. Nomura, B. Yang, H. Shimotani, R. Suzuki, M. Toh, C. Kloc, X. D. Xu, R. Arita, N. Nagaosa, and Y. Iwasa, *Nat. Phys.* **9**, 563 (2013).

²⁴B. Fallahazad, H. C. Movva, K. Kim, S. Larentis, T. Taniguchi, K. Watanabe, S. K. Banerjee, and E. Tutuc, *Phys. Rev. Lett.* **116**, 86601 (2016).

²⁵J. M. Riley, W. Meevasana, L. Bawden, M. Asakawa, T. Takayama, T. Eknapakul, T. K. Kim, M. Hoesch, S. Mo, H. Takagi, T. Sasagawa, M. S. Bahramy, and P. D. C. King, *Nat. Nanotechnol.* **10**, 1043 (2015).

²⁶J. Suh, T. Park, D. Lin, D. Fu, J. Park, H. J. Jung, Y. B. Chen, C. Ko, C. Jang, Y. H. Sun, R. Sinclair, J. Chang, S. Tongay, and J. Q. Wu, *Nano Lett.* **14**, 6976 (2014).

²⁷Y. Jin, D. H. Keum, S. J. An, J. Kim, H. S. Lee, and Y. H. Lee, *Adv. Mater.* **27**, 5534 (2015).

²⁸J. Yang, Y. C. Liu, H. M. Lin, and C. C. Chen, *Adv. Mater.* **16**, 713 (2004).

²⁹M. Mikami, T. Eto, J. F. Wang, Y. Masa, and M. Isshiki, *J. Cryst. Growth* **276**, 389 (2005).

³⁰M. Bougouma, B. Guel, T. Segato, J. B. Legma, and M. D. Ogletree, *Bull. Chem. Soc. Ethiop.* **22**, 225 (2008).

³¹X. Luo, Y. Y. Zhao, J. Zhang, M. L. Toh, C. Kloc, Q. H. Xiong, and S. Y. Quek, *Phys. Rev. B* **88**, 195313 (2013).

³²E. Del Corro, H. Terrones, A. Elias, C. Fantini, S. Feng, M. A. Nguyen, T. E. Mallouk, M. Terrones, and M. A. Pimenta, *ACS Nano* **8**, 9629 (2014).

³³B. Chakraborty, A. Bera, D. Muthu, S. Bhowmick, U. V. Waghmare, and A. K. Sood, *Phys. Rev. B* **85**, 161403 (2012).

³⁴H. Qiu, T. Xu, Z. Wang, W. Ren, H. Nan, Z. Ni, Q. Chen, S. Yuan, F. Miao, F. Song, G. Long, Y. Shi, L. Sun, J. Wang, and X. Wang, *Nat. Commun.* **4**, 2642 (2013).

³⁵B. Radisavljevic and A. Kis, *Nat. Mater.* **12**, 815 (2013).

³⁶T. C. Banwell and A. Jayakumar, *Electron. Lett.* **36**, 291 (2000).

³⁷D. J. Groenendijk, M. Buscema, G. A. Steele, S. Michaelis De Vasconcellos, R. Bratschitsch, H. S. van der Zant, and A. Castellanos-Gomez, *Nano Lett.* **14**, 5846 (2014).

³⁸M. S. Long, E. F. Liu, P. Wang, A. Y. Gao, H. Xia, W. Luo, B. G. Wang, J. W. Zeng, Y. J. Fu, K. Xu, W. Zhou, Y. Y. Lv, S. H. Yao, M. H. Lu, Y. F. Chen, Z. H. Ni, Y. M. You, X. A. Zhang, S. Q. Qin, Y. Shi, W. D. Hu, D. Y. Xing, and F. Miao, *Nano Lett.* **16**, 2254 (2016).

³⁹L. Britnell, R. V. Gorbachev, R. Jalil, B. D. Belle, F. Schedin, A. Mishchenko, T. Georgiou, M. I. Katsnelson, L. Eaves, S. V. Morozov, N. M. R. Peres, J. Leist, A. K. Geim, K. S. Novoselov, and L. A. Ponomarenko, *Science* **335**, 947 (2012).

⁴⁰C. R. Dean, A. F. Young, I. Meric, C. Lee, L. Wang, S. Sorgenfrei, K. Watanabe, T. Taniguchi, P. Kim, K. L. Shepard, and J. Hone, *Nat. Nanotechnol.* **5**, 722 (2010).